Two Types of Right Dislocation in Korean*

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1 Introduction

While the left periphery has been paid great attention in syntactic theory, the right periphery remains relatively unexplored. This paper aims to contribute to the understanding of the right periphery, focusing on right dislocation (henceforth, RD) in Korean.

In the canonical order of a sentence in Korean, the subject and the object precede the verb, as in example (1a). Korean is a scrambling language, and the surface word order is relatively free as a result of scrambling. RD involves an argument to the right of the finite verb, as exemplified in (1b) and (1c) (Post-verbal elements are underlined hereafter).

(1) a.	iyowen-i	yuinna-lul	miwehayyo.	(SOV: canonical order)
	L.-NOM	Y.-ACC	hates	
b.	iyowen-i	miwehayyo	<u>yuinna-lul</u>.	(SVO: RD of the object)
	L.-NOM	hates	Y.-ACC	
c.	yuinna-lul	miwehayyo	<u>iyowen-i</u>.	(OVS: RD of the subject)
	Y.-ACC	hates	L.-NOM	

'Lee Yowon hates Yu Inna.'

RD is observed only in colloquial speech of Korean, and all example sentences in this paper end in a colloquial ending such as *-yo* above. There is no restriction of the number of right-dislocated elements in principle, as in the case of leftward scrambling, and more than one elements can be located post-verbally. The same words contained in the sentence in (1a) can also be ordered as in (2), in which both the subject and the object follow the verb. In ditransitive sentences as in (3), all three arguments can be located in the post-verbal position as in (3e), as well as any two of the three arguments as in (3b), (3c), and (3d).

(2) miwehayyo	<u>iyowen-i</u>	<u>yuinna-lul</u>
hates	L.-NOM	Y.-ACC

'Lee Yowon hates Yu Inna.'

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(3) a. iyowen-i inna-eykey yangmal-ul cwuesseyo
L.-NOM Inna-DAT socks-ACC gave
b. yangmal-ul cwuesseyo iyowen-i inna-eykey
socks-ACC gave L.-NOM Inna-DAT
c. iyowen-i cwuesseyo inna-eykey yangmal-ul
L.-NOM gave Inna-DAT socks-ACC
d. inna-eykey cwuesseyo iyowen-i yangmal-ul
Inna-DAT gave L.-NOM socks-ACC
e. cwuesseyo iyowen-i inna-eykey yangmal-ul
gave L.-NOM Inna-DAT socks-ACC
'Lee Yowon gave Inna socks.'

Previous studies on Korean RD have assumed that it involves some kind of movement, and there have been two branches of the movement analysis of RD in Korean. One approach has assumed that RD derives from rightward movement of the post-verbal element, as illustrated in (4a) (Choe 1987, Choi 2008, Ko and Choi 2009). The other position has argued that RD involves a bi-clausal structure and the post-verbal argument results from ellipsis of the following material, as shown in (4b) (e.g., Chung 2012, Park and Kim 2009, Yim 2013).

(4) a. $[_{CP}$ [… t_i verb] RD-phrase$_i$]
b. $[_{CP}$ … *pro*$_i$ verb] $[_{CP}$ RD-phrase$_i$ ~~[… t_i verb]~~]

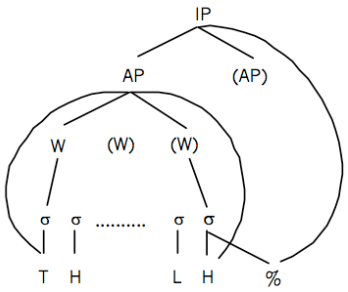
By investigating prosodic and information structure of Korean RD, as well as its syntactic structure, this paper proposes that there are two different types of RD in Korean. In particular, one is a mono-clausal structure involving rightward movement as in (4a) (henceforth, RD1), and the other is a bi-clausal structure with leftward movement and ellipsis as in (4b) (henceforth, RD2). In the rest of this paper, sections 2, 3, and 4 will be devoted to prosodic structure, information structure, and syntactic structure of the two types of RD in Korean respectively, and section 5 will summarizes and concludes the paper.

2 Prosodic Structure

This section deals with prosodic realizations of RD in Korean. In this section, Korean is limited to the Seoul dialect, and other dialects in Korean may show different intonation patterns, although the essential part of RD prosody would be shared.

It may be useful to start out by introducing preliminaries of Seoul Korean prosody. According to Jun (1993), there are two intonationally-defined prosodic units in Seoul Korean: Intonational Phrase (IP) and Accentual Phrase (AP). These are schematically represented in (5).

(5) Intonational Structure of Seoul Korean (Jun 2000)



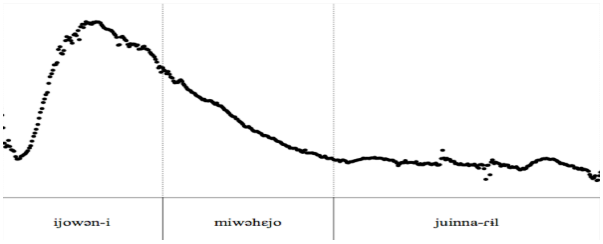
An AP, a smaller unit, can have more than one word and is realized with a sequence of phrasal tones, LHLH, except when the initial consonant of the AP is either a tense or an aspirated stop, which is realized with HHLH. An IP, marked by a boundary tone and final lengthening on the IP-final syllable, consists of one or more APs. There are at least nine boundary tones in Seoul Korean: L%, H%, LH%, HL%, LHL%, HLH%, HLHL%, LHLH%, LHLHL%. Among these, L% is commonly used in a declarative sentence. The boundary tones preempt the AP-final rising tone.

As for the prosody of contrastive focus, Jun and Lee (1998) experimentally showed that contrastive focus always initiated an AP, and focused F0 peaks were relatively higher and longer than non-focused peaks. In addition, the pitch range of the following words was considerably lowered, and dephrasing of the post-focal items was optional.

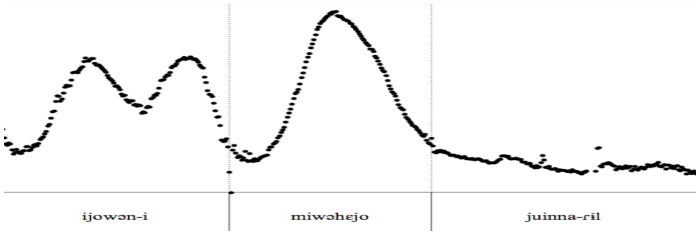
Looking at the RD construction, two different prosodic patterns are observed. RD1 involves a high focus peak, and dephrasing follows it. The figure in (6) shows a pitch track of subject focus reading of the sentence in (1b), and the figure in (7) shows a pitch track of verb focus reading of the same sentence. All figures of pitch movement in this paper are extracted from recordings from a female native speaker of Seoul Korean.

Both in (6) and (7), post-focal phrase(s) are completely reduced in prosodic realization. To be specific, the post-focal material that contains the right-dislocated element cannot have an F0 peak or form its own AP, but is always incorporated into the preceding AP. Also, they are significantly reduced in amplitude. This dephrasing is one of the typical characteristics of focus prosody in Korean, as mentioned earlier, although it is optional in the normal word order. The fact that RD1 shows an extreme case of post-focal prosodic reduction might indicate that there is a constraint that requires post-verbal arguments in verb-final languages to be not prominent in prosodic realization.

(6) RD1: Subject focus reading of (1b)

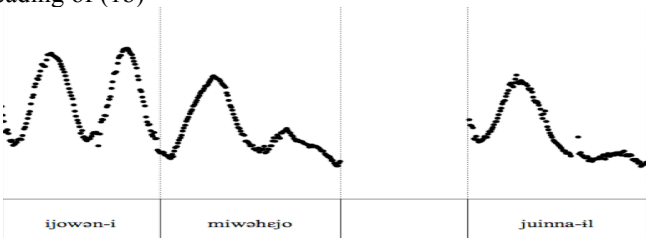


(7) RD1: Verb focus reading of (1b)



In contrast, the prosody of RD2 shows two IPs, unlike the first type having a single IP. The sentence in (1b) can also be read as illustrated in (8).

(8) RD2 reading of (1b)



Here the subject *iyowen-i* and the verb *miwehayyo* form the first IP, and the right-dislocated object *yuinna-lul* forms the second IP, following a pause. Specifically, a boundary tone and final lengthening occur twice, i.e., at the end of the verb and at the end of the right-dislocated element. Unlike the first AP on the subject, which we can clearly see the LHLH tonal pattern, the second F0 peak of the second AP on the verb has a much lower peak than others due to the boundary tone, HL%. The third AP on the right-dislocated object also shows the LHLH tonal pattern, but again, the second F0 peak is very low, influenced by the boundary tone, HL%. Although in this pitch track it is hard to see the IP-final

lengthening, there seem to be the final lengthening effects in both of the IPs, which will need future investigation.

In sum, there are two very different prosodic patterns of Korean RD. One involves focus prosody in a single IP, with the right-dislocated phrase totally deaccented, and the other involves two IPs, one of which is formed on the right-dislocated element.

3 Information Structure

Each of the two different prosodic patterns of RD in Korean is used in a very different context, and they are almost in complementary distribution. This section investigates their information structures. To be specific, I argue that the right-dislocated element in RD1 always conveys given information, whereas the right-dislocated one in RD2 is likely to convey new information. Here I adopt Schwarzschild's (1999) definition of givenness, which can be informally stated that a given constituent is entailed by the previous discourse. So I consider a constituent given iff it has been mentioned or at least entailed in the discourse previously. I also follow Schwarzschild's (1999) view that what is not given is all considered new information.

Let us consider the context of broad focus, as in (9). Here each component of speaker B's answer carries new information, and sentences involving RD as in (9c) and (9d) with RD1 prosody sound awkward as an answer to the question in (9a), whereas the sentence in (9b) with the basic word order is totally fine. In contrast, reading the same sentences (9c) and (9d) with the prosody of RD2 makes the answer acceptable. I attribute this contrast to the givenness of the right-dislocated element. That is, RD1 cannot occur in broad focus, as it cannot convey new information on the post-verbal element, whereas RD2 can, as it can carry new information post-verbally.

(9) Broad focus

a. A: mwusun il-iya?
what matter-be
'What is the matter?'

b. B: mina-ka nana-lul ttaylyesse.
Mina-NOM Nana-ACC hit

c. B: mina-ka ttaylyesse nana-lul. (#RD1, ^{OK}RD2)
Mina-NOM hit Nana-ACC

d. B: nana-lul ttaylyesse mina-ka. (#RD1, ^{OK}RD2)
Nana-ACC hit Mina-NOM
'Mina hit Nana.'

Meanwhile, there are previous studies on RD that have suggested that it is inherently focused elements that cannot undergo RD (without distinction between RD1 and RD2). Choe (1987) mentioned that wh-phrases and phrases that contain a wh-phrase could not be right-dislocated, as shown in (10). Choi (2008) somewhat broadens Choe's (1987) generalization, assuming that inherently focused elements including wh-phrases cannot undergo RD, pointing out that an answer to a question cannot be located post-verbally, as shown in (11).

(10) chelswu-ka poasssupnikka mwues-ul?
Chelswu-NOM see-PST-Q what-ACC
'What did Chelswu see?' (Choe 1987, p. 43; underline added)

(11) a. A: nwu-ka nayngmyen-ul sikhy-ess-ni?
who-NOM cold.noodles-ACC order-PST-Q
'Who ordered cold noodles?'

b. #B: nayngmyen-ul sikhy-ess-e sangcheli-ka
cold.noodles.ACC order-PST-DC Sangchel-NOM
'Sangchel ordered cold noodles.'
(Choi 2008, p. 65; underline added)

Considering their syntactic analyses, it seems that RD dealt with in Choe (1987) and Choi (2008) corresponds to RD1 in this paper. Notice, however, that this generalization on RD1 in the previous studies is not general enough to explain the restrictions on information structure of RD1. As seen earlier, in broad focus, the right-dislocated phrase in RD1 is not allowed, even though it is not inherently focused. Rather, both wh-phrases and answers to a question involve new information, not previously mentioned, and the current givenness account can explain their inability to undergo RD1, as well as no RD1 in broad focus.

The following examples in (12) confirm that it is givenness, not focus, that regulates RD1. Speaker A says the sentence out of the blue, which indicates that every single element in the sentence carries new information, and RD1 of the focused object is not possible in (12b). The focused phrase, however, can be right-dislocated in (12c), given that the sentence is a contradiction of the assertion in (12a) and the focused phrase is given. This shows that only elements with given information can undergo RD1, regardless of whether it is focused or not.

(12) a. A: mina-ka mwul-man masyesseyo.
Mina-NOM water-only drank

b. A: mina-ka masyesseyo mwul-man. (#RD1, ^{OK}RD2)
Mina-NOM drank water-only
'Mina drank water only.'

c. B: (aniyo) nana-ka masyesseyo mwul-man. (^{OK}RD1, ^{OK}RD2)
no Nana-NOM drank water-only
'(No,) Nana drank water only.'

In addition, it is most likely to be the case that there is a narrowly-focused element in the non-dislocated part of the RD1 sentence, as shown in the pitch track previously. The sentence in (1b) has focus on a non-dislocated phrase, either on the subject as in (6) or on the verb as in (7).

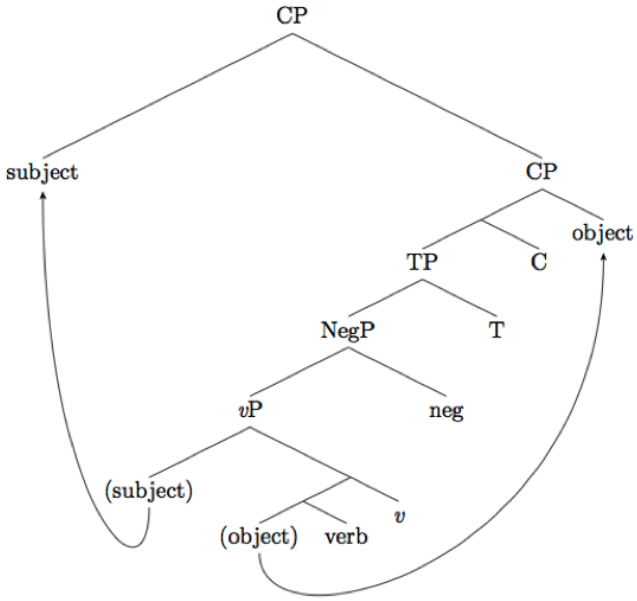
In contrast with RD1, RD2 allows RD of a phrase that conveys new information, as seen above in (9) and (12), and does not need focus. It appears that post-verbal phrases in RD2 tend to involve new information and sometimes have given information as in (12c). To make sure of this, a spoken corpus study would be necessary, which I leave to future research.

To summarize, the right-dislocated phrase in RD1 must convey given information. Since there is a narrowly-focused element preceding the post-verbal argument, this may indicate that items that comes after the verb in the verb-final language are required not to be salient in information structure, as well as in its prosodic realization. In contrast, RD2 does not have strict restrictions on the information structure of the post-verbal argument, and tends to have new information on the post-verbal phrase, while it seems that it is not impossible to carry given information.

4 Syntactic Structure

This section investigates syntactic structures of the two types of RD in Korean, which are mapped to the two different prosodic and information structures discussed above. First, I argue that rightward movement in a single clause derives RD1, and the post-verbal argument is adjoined at the CP level in Korean, as in Turkish (Kural 1997). A brief structure for an SVO sentence with RD1 is illustrated in (13). Recall that RD1 involves narrow focus in a phrase preceding the post-verbal one, and here the subject that carries the focus undergoes focus movement to the Spec of the CP, which is higher than the post-verbal object.

(13) Structure of RD1 (SVO with subject focus)



On the other hand, RD2 involves leftward movement of the post-verbal phrase and ellipsis of the following phrases in a bi-clausal construction. To be specific, in RD2, two (nearly) identical clauses are first juxtaposed as in (14a), and the second clause undergoes leftward scrambling of the phrase that is post-verbal on the surface, as in (14b). And the rest of the second clause deletes under identity, as in (14c).

(14) Structure of RD2 (SVO)[1]

a. $[_{CP1}$ S *pro*$_i$ V $]$ $[_{CP2}$ S O_i V $]$
b. $[_{CP1}$ S *pro*$_i$ V $]$ $[_{CP2}$ O_i [S t_i V]]
c. $[_{CP1}$ S *pro*$_i$ V $]$ $[_{CP2}$ O_i ~~[S t_i V]~~]

Evidence for these two structures is presented in the following. First, both of the structures involve movement of the post-verbal phrase, because both RD1 and RD2 are island-sensitive. For example, the sentence in (15) involves a wh-island, and is ungrammatical with both RD1 and RD2 prosodies.

[1] For the empty category in the first clause, there have been several different proposals (see, for example, Park and Kim 2009). This paper uses *pro*, for simplicity, and does not discuss the nature of this empty category.

(15) *[t mwues-ul sassnun-ci] chelswu-ka molunta yenghuy-ka
what-ACC bought-COMP Chelswu-NOM don't.know Yenghuy-NOM
'Chelswu doesn't know what Yenghuy bought.'
(Choe 1987, p. 43; underline added)

Although both involve movement, RD1 consists of one clause and RD2 consists of two clauses. The sentence in (16) shows this contrast. This sentence is totally bad with RD1 prosody but becomes grammatical with RD2 prosody. I argue that this is because RD1 and RD2 have different syntactic structures. In RD1, the post-verbal NP *nana-lul* 'Nana-ACC' and its co-indexed pronoun *kunye-lul* 'she-ACC' are in the same clause, and thus violate Principle B. In RD2, in contrast, the post-verbal NP and its co-indexed pronoun stay in separate clauses, and are free from the binding condition, resulting in the grammaticality. It may also support that, in RD2, there is a correlate of the right-dislocated element in the first clause, which is overt in (16) or covert in the other earlier examples.

(16) mina-ka kunye-lul$_i$ miwehayyo nana-lul$_i$
Mina-NOM she-ACC hates Nana-ACC
'Mina hates Nana.' (*RD1, ^{OK}RD2)

Another syntactic difference between RD1 and RD2 is found when involving right-dislocated pre-nominal modifiers. Interestingly, pre-nominal modifiers, which cannot undergo movement in Korean, can be right-dislocated in RD2, but not in RD1. In (17), the pre-nominal modifier *acwu yeppun* 'very pretty' is located post-verbally, and this sentence is good only when read with the prosody of RD2, not of RD1. This may be because, in RD1, it is the pre-nominal modifier by itself that has to move to the right of the verb, and the movement is blocked due to the restriction on the syntactic movement in Korean. In RD2, however, the appearance of a non-movable post-verbal element can be explained by the movement of the whole NP, *acwu yeppun sinpwu-lul* 'very pretty bride-ACC' and the deletion of the nominal *sinpwu-lul* 'bride-ACC' after the movement, under identity to the one in the first clause, as demonstrated in (18).

(17) John-i sinpwu-lul manna-ss-ta acwu yeppun.
John-NOM bride-ACC meet-PST-DC very pretty
'John met a bride, very pretty.'
(Park and Kim 2009, p. 26; underline added)

(18) [$_{CP1}$ John-i sinpwu-lul manna-ss-ta]
[$_{CP2}$ acwu yeppun ~~sinpwu-lul$_i$~~ ~~John-i t$_i$ manna-ss-ta~~]

The two types RD also show differences in scope interpretations. First, in a sentence that involves a universal quantifier and negation as in (19), the right-dislocated quantifier takes rigid scope over negation without reconstruction in RD1. This shows that the landing site of the right-dislocated phrase is higher than negation[2]. On the other hand, RD2 reading of the same sentence shows ambiguous scope interpretations between the universal quantifier and negation. It can be explained by the bi-clausal structure; when the right-dislocated phrase is interpreted in the *pro* position in the first clause (cf. (14)), the sentence gets the wide scope reading of the universal quantifier with respect to negation, and when it is interpreted in the surface position in the second clause, it outscopes negation.

(19) mina-ka ilkci-anh-ass-e motun chayk-ul
Mina-NOM read-NEG-PST-DC all book-ACC
'Mina didn't read all books.'
(RD1: $^{OK}\forall > \neg$, $^{*}\neg > \forall$; RD2: $^{OK}\forall > \neg$, $^{OK}\neg > \forall$)

Doubly-quantified sentences show a similar scope difference between RD1 and RD2. In the sentence in (20), which involves an existential quantifier subject and a right-dislocated universal quantifier object, RD1 only allows the surface scope reading[3], whereas RD2 allows ambiguous scope interpretations. As illustrated in (13), the subject, which is likely to be focused here, undergoes focus movement to a higher position, [Spec,CP], than the right-dislocated object, resulting in the wide scope of the subject with respect to the object. In RD2, in contrast, the scope interpretation depend on whether the post-verbal quantifier is interpreted in the surface position in the second clause, where it takes scope over the subject, or in the *pro* position, where it takes scope under the subject. Thus, ambiguous scope interpretations are possible in RD2.

(20) nwukwunka-ka miwehayyo motun yeca-lul
someone-NOM hates all woman-ACC
'Someone hates all women.' (RD1: , RD2:)
(RD1: $^{OK}\exists > \forall$, $^{*}\forall > \exists$; RD2: $^{OK}\exists > \forall$, $^{OK}\forall > \exists$)

So far I have argued for two different syntactic structures for Korean RD: a mono-clausal structure involving rightward movement to the CP-adjoined position, and a bi-clausal structure involving leftward movement and ellipsis. For

[2] Unlike the case of leftward scrambling, rightward movement in RD1 does not seem to allow reconstruction.

[3] In RD1, a right-dislocated argument always scopes under a preceding argument, and thus if right-dislocated, the subject is under the scope of the object. For more data and analysis, see Yun (2013).

RD1, why does the landing site of RD have to be the CP-adjoined position? The example in (19) indicates that the right-dislocated phrase is located higher than negation, but there are several possible landing cites in the right periphery other than a CP-adjoined position, such as a TP-adjoined position. Evidence for the CP-adjoined position is first coming from the fact that a right-dislocated phrase is always preceded by the C position. In (21), for example, the right-dislocated object follows the Q morpheme *-ni*, which is assumed to stay in the C position. Also, morpheme order in Korean verbal complex may show that there is no position for the right-dislocated phrase other than the CP-adjoined position. In (21), the past morpheme *-yess* cannot be separate from the preceding verbal root *miweha-* 'to hate' or the following Q morpheme *-ni*. Assuming that the tense morpheme stays in the T and the Q morpheme stays in the C position, we would not get the correct word order, if the right-dislocated phrase were located in the TP-adjoined position, because the right-dislocated phrase would stay after the past morpheme *-yess* but before the Q morpheme *-ni*, resulting in **miweha-yess nana-lul-ni*.

(21) mina-ka miweha-yess-ni nana-lul?
Mina-NOM hates-PST-Q Nana-ACC
'Does Mina hate Nana?'

Previously, Choi (2008) and Ko and Choi (2009) proposed that the right-dislocated phrase moves to the inner edge of *v*P, based on the scope data in which the non-dislocated subject outscopes the right-dislocated object, as shown in (20). It appears that RD in the studies refers to RD1 in this paper, and the *v*P-adjunction analysis cannot cover all data relevant to RD1 presented here. First, right-adjunction under *v*P cannot explain the wide scope of the right-dislocated phrase over negation, as shown in (19), because the right-dislocated phrase always stays lower than negation. Also, under this analysis, merge of the subject has to happen after RD, and thus RD of a subject, as in the sentence in (1c), is basically impossible. This is because the subject is merged into the Spec of the *v*P, and RD of the subject at the *v*P level would allow only the right-adjunction to the *outer* edge of *v*P. Therefore, the subject RD is not possible as long as they assume that the site of RD is the *inner* edge of *v*P, which is lower than the Spec of the *v*P. For more critical review of the previous work on RD, see Yun (2013).

As mentioned in section 1, previous approaches to RD in Korean have assumed that there exists only one RD construction, and argued either for the rightward movement or for the bi-clausal adjunction and leftward movement. As shown in this paper, however, depending on the prosody and information structure, sentences showing RD are grammatical or not, and their scope interpretations differ. This may imply that there are actually two different syntactic structures for RD, which has been regarded as a single construction,

and each structure is mapped to the different prosodic and information structure discussed earlier.

5 Conclusion

In this paper we have explored prosodic structure, information structure, and syntactic structure of RD in Korean, and showed that there are two different types of RD in Korean. Prosodically, RD1 shows remarkable prosodic reduction in the post-verbal element, whereas the post-verbal phrase in RD2 forms its own AP and IP. For information structure, RD1 carries given information, and RD2 tends to carry new information. Syntactically, RD1 involves a mono-clausal construction that undergoes rightward movement or the post-verbal phrase to the CP-adjoined position, whereas RD2 involves a bi-clausal construction that undergoes leftward movement of the post-verbal phrase and ellipsis of the following phrases under identity.

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